

BSDS (2025 Semester II): Statistics II

Quiz 2

NAME :	ROLL :
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Time: 60 minutes

Total marks: 30

1. Let $Z, X_1, \dots, X_n$ be i.i.d. samples from `normal`(0, 1) distribution. Consider the random variables

$$S_n = \frac{X_1^2 + \dots + X_n^2}{n}, \quad \text{and} \quad T_n = \frac{Z}{\sqrt{S_n}}.$$

- (a) What is the distribution of T_n ? Justify your answer.

[4]

(b) Identify the constant c , such that $S_n \xrightarrow{p} c$ as $n \rightarrow \infty$. Justify your answer.

[4]

- (c) Hence or otherwise show, with proper justification, that $T_n \xrightarrow{d} W$ as $n \rightarrow \infty$, where $W \sim \text{normal}(0, 1)$. [4]

2. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{normal}(\mu, \sigma^2)$, where σ^2 is unknown. Consider the problem of testing $H_0 : \mu = 0$ against $H_1 : \mu > 0$. Consider the test function of the form

$$\phi^*(\mathbf{x}) = \begin{cases} 1 & \text{if } t_n \geq c_\alpha, \\ 0 & \text{otherwise,} \end{cases}$$

where t_n is a realization of the statistic

$$T_n = \frac{\sqrt{n}\bar{X}_n}{S_n}, \quad \text{and} \quad S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

- (a) Find appropriate scaling factors $\kappa_1(\sigma, n)$ and $\kappa_2(\sigma, n)$ such that the distributions of $\bar{X}_n/\kappa_1(\sigma, n)$ and $S_n^2/\kappa_2(\sigma, n)$ are $\text{normal}(0, 1)$ and χ_r^2 , respectively, under H_0 . Also, indicate the appropriate degrees of freedom ‘ r ’.

[2 + 2]

(b) Using part (a) or otherwise find the exact distribution of T_n under H_0 .

[3]

- (c) Find the size of the test ϕ^* . Hence, obtain the appropriate choice of c_α such that the size of ϕ^* is equal to α . [5]

(d) Based on Q.1, can you find the asymptotic distribution of T_n under H_0 ?

[3]

- (e) How would the cutoff c_α change if the asymptotic distribution of T_n is used instead of the exact distribution? [3]